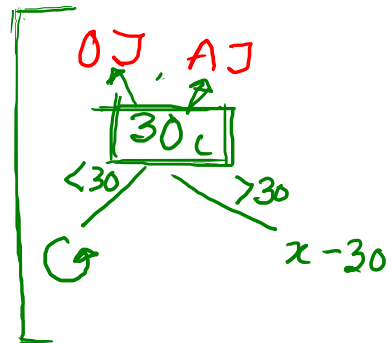


13.2 Finite-State Machines with Output

Before giving formal definitions, we will show how a vending machine can be modeled. A vending machine accepts nickels (5 cents), dimes (10 cents), and quarters (25 cents). When a total of 30 cents or more has been deposited, the machine immediately returns the amount in excess of 30 cents. When 30 cents has been deposited and any excess refunded, the customer can push an orange button and receive an orange juice or push a red button and receive an apple juice. We can describe how the machine works by specifying its states, how it changes states when input is received, and the output that is produced for every combination of input and current state.

The machine can be in any of seven different states $s_i, i = 0, 1, 2, \dots, 6$, where s_i is the state where the machine has collected $5i$ cents. The machine starts in state s_0 , with 0 cents received. The possible inputs are 5 cents, 10 cents, 25 cents, the orange button (O), and the red button (R). The possible outputs are nothing (n), 5 cents, 10 cents, 15 cents, 20 cents, 25 cents, an orange juice, and an apple juice.

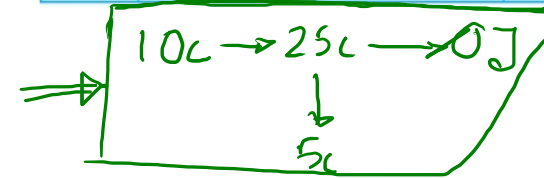
We illustrate how this model of the machine works with this example. Suppose that a student puts in a dime followed by a quarter, receives 5 cents back, and then pushes the orange button for an orange juice. The machine starts in state s_0 . The first input is 10 cents, which changes the state of the machine to s_2 and gives no output. The second input is 25 cents. This changes the state from s_2 to s_6 , and gives 5 cents as output. The next input is the orange button, which changes the state from s_6 back to s_0 (because the machine returns to the start state) and gives an orange juice as its output.



$s_0 \rightarrow 0 \rightarrow 0$
 $s_0 \rightarrow 5 \times 0 \rightarrow 0c$
 $s_1 \rightarrow 5 \times 1 \rightarrow 5c$
 $s_6 \rightarrow 5 \times 6 \rightarrow 30c$

TABLE 1 State Table for a Vending Machine.

State	Next State					Output				
	Input 5	Input 10	Input 25	O	R	Input 5	Input 10	Input 25	O	R
s_0	s_1	s_2	s_5	s_0	s_0	n	n	n	n	n
s_1	s_2	s_3	s_6	s_1	s_1	n	n	n	n	n
s_2	s_3	s_4	s_6	s_2	s_2	n	n	5	n	n
s_3	s_4	s_5	s_6	s_3	s_3	n	n	10	n	n
s_4	s_5	s_6	s_6	s_4	s_4	n	n	15	n	n
s_5	s_6	s_6	s_6	s_5	s_5	n	5	20	n	n
s_6	s_6	s_6	s_6	s_0	s_0	5	10	25	OJ	AJ



$5 \times 6 \rightarrow 30c$

$15 \rightarrow 15c \text{ short}$
 $s_5 \rightarrow 25 \rightarrow 10c \text{ excess}$
 $s_6 \rightarrow 40c \rightarrow \text{Return } 10c$
 $\rightarrow \text{get OJ/AJ on pressing}$

Deterministic $\rightarrow$
 Non-deterministic $\rightarrow$
 Finite State Automata.

DEFINITION 1

A finite-state machine $M = (S, I, O, f, g, s_0)$ consists of a finite set S of states, a finite input alphabet I , a finite output alphabet O , a transition function f that assigns to each state and input pair a new state, an output function g that assigns to each state and input pair an output, and an initial state s_0 .

~~10c~~ → ~~25c~~ → ~~10c~~ → 0J
 $\downarrow$
5c
 10 → 10 → 10 → RJ

TABLE 1 State Table for a Vending Machine.

State	Next State					Output				
	5	10	25	O	R	5	10	25	O	R
s ₀	s ₂	s ₂	s ₅	s ₀	s ₀	n	n	n	n	n
s ₁	s ₂	s ₂	s ₆	s ₁	s ₁	n	n	n	n	n
s ₂	s ₂	s ₂	s ₆	s ₂	s ₂	n	n	5	n	n
s ₃	s ₄	s ₅	s ₆	s ₃	s ₃	n	n	10	n	n
s ₄	s ₅	s ₆	s ₆	s ₄	s ₄	n	n	15	n	n
s ₅	s ₆	s ₆	s ₆	s ₅	s ₅	n	5	20	n	n
s ₆	s ₆	s ₆	s ₆	s ₀	s ₀	5	10	25	OJ	AJ

→ $f(s_i, I) \rightarrow s_{i+1}$
 $g(s_i, I) \rightarrow O$

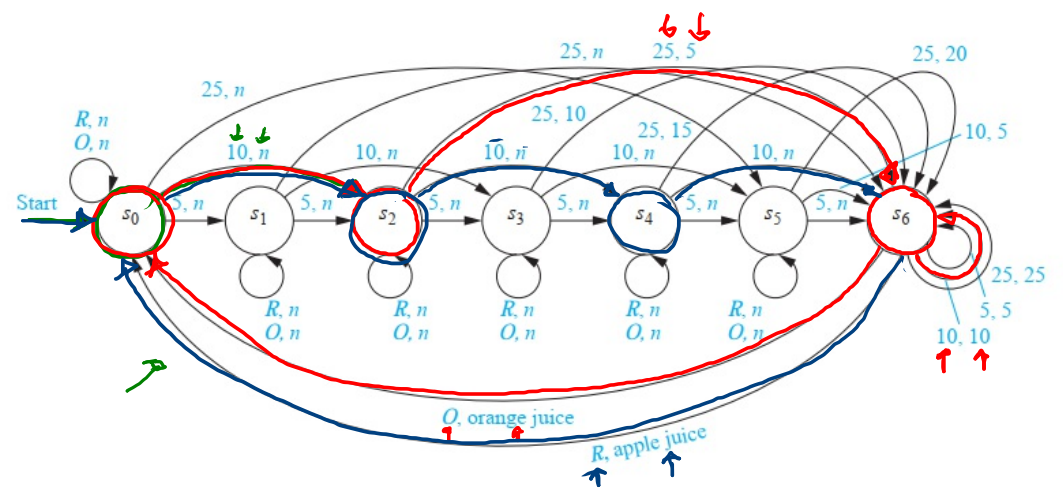
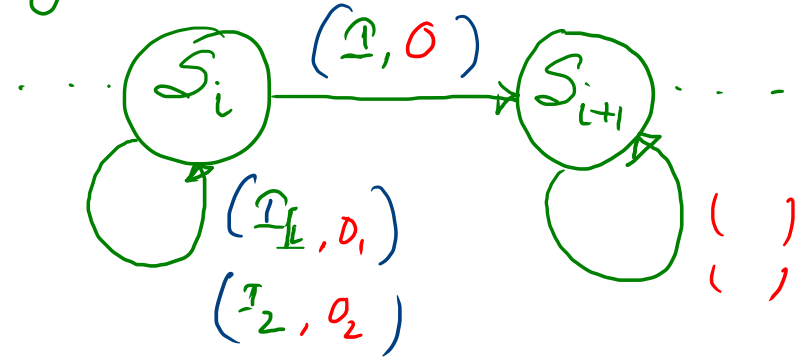


FIGURE 1 A Vending Machine.

Example 2: Fill Up the state table of the following FSM.

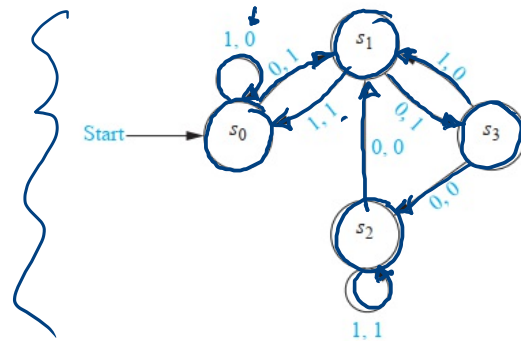
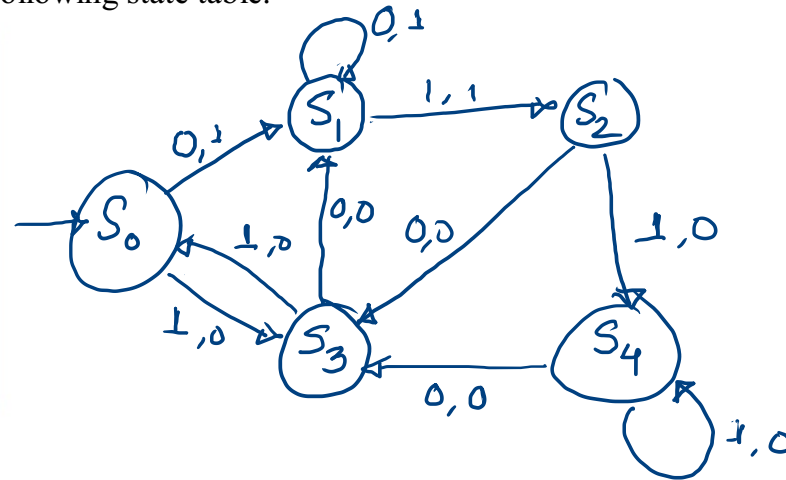


TABLE 2				
State	f		g	
	Input		Input	
	0	1	0	1
s_0	s_1	s_0	1	0
s_1	s_3	s_0	1	1
s_2	s_1	s_2	0	1
s_3	s_2	s_1	0	0

Example 3: Draw the FSM for the following state table.

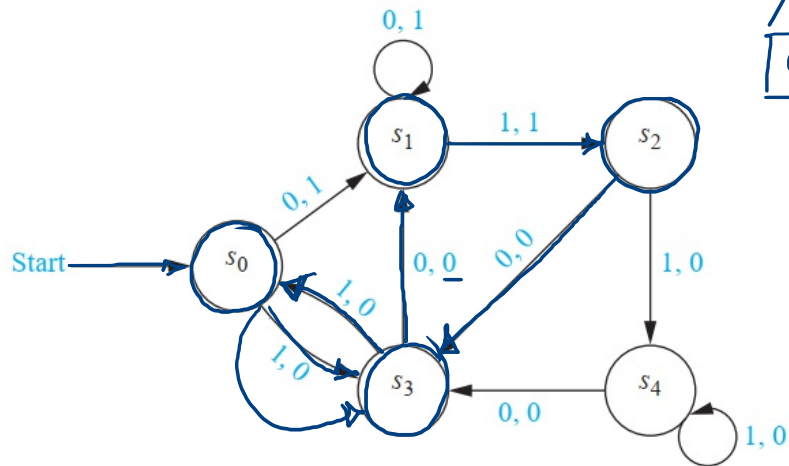
TABLE 3				
State	f		g	
	Input		Input	
	0	1	0	1
s_0	s_1	s_3	1	0
s_1	s_1	s_2	1	1
s_2	s_3	s_4	0	0
s_3	s_1	s_0	0	0
s_4	s_3	s_4	0	0



An input string takes the starting state through a sequence of states, as determined by the transition function. As we read the input string symbol by symbol (from left to right), each input symbol takes the machine from one state to another. Because each transition produces an output, an input string also produces an output string.

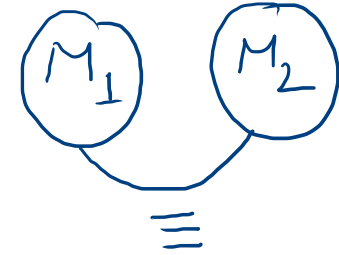
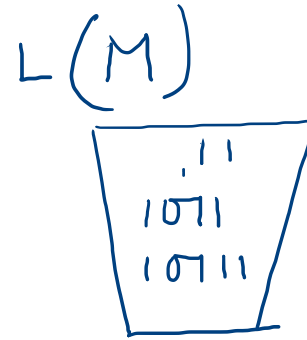
Example 4:

Find the output string generated by the finite-state machine in the FSM if the input string is 101011.



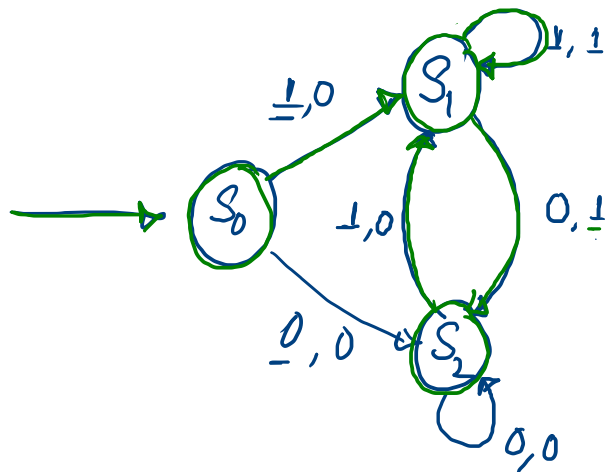
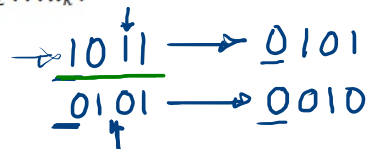
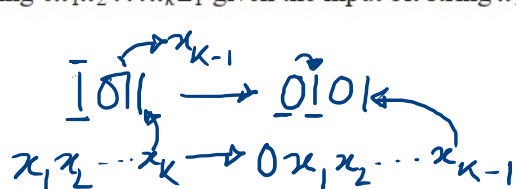
$L \longrightarrow R$

~~101011~~
 001000



Example 5:

- An important element in many electronic devices is a unit-delay machine, which produces as output the input string delayed by a specified amount of time. How can a finite-state machine be constructed that delays an input string by one unit of time, that is, produces as output the bit string $0x_1x_2 \dots x_{k-1}$ given the input bit string $x_1x_2 \dots x_k$?



$S_0 \xrightarrow{1} 0$

$S_1 \xrightarrow{10} 01$

$S_2 \xrightarrow{101} 010$

$S_1 \xrightarrow{1011} 0101$

$S_1 \xrightarrow{1011} \underline{0}101$